Project Introduction

Notes

- 50 % Projet
  - Rapport final
  - Fichiers Altium
  - Cahier de charge
  - Cahier de tests
  - Malus cahier de labo
- 50 % Examen dernier semaine de semestre

Plannification: Documents à fournir

Week	Mercredi		Vendredi	
1	2026-02-18	Cahier de charge	2026-02-20	Cahier de labo
2	2026-02-25		2026-02-27	Cahier de labo
3	2026-03-04	Cahier de labo	2026-03-06	Cahier de labo
4	2026-03-11		2026-03-13	
5	2026-03-18	Cahier de labo	2026-03-20	Cahier de labo
6	2026-03-25		2026-03-27	PCB fabrication docs
7	2026-04-01	Cahier de tests	2026-04-03	
Paques	2026-04-08		2026-04-10	
8	2026-04-15		2026-04-17	
9	2026-04-22	PCB solder (no doc)	2026-04-24	Cahier de labo
10	2026-04-29		2026-05-01	Cahier de labo
11	2026-05-06	Cahier de labo	2026-05-08	
Fin project				

Rapport final: 19/05/26 08h00

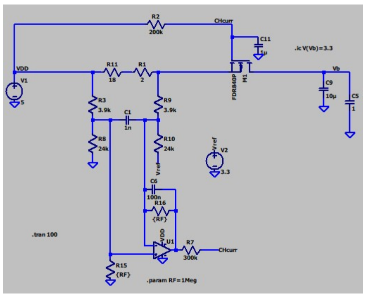
Cahier de labo (Cyberlearn)

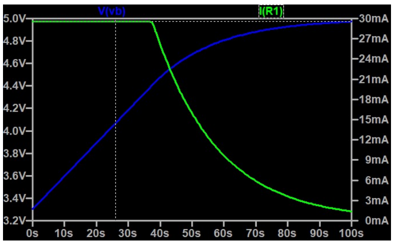
- Un carnet à tenir à jour pendant le projet
- Quesque vous avez fait ?
- C'était quoi les résultats ?
- Informelle!
- Pas des exigences sur des graphiques
  - Mais: Avec chemin + nom de fichier pour le reproduir
- Notes:
  - 0: Insuffisant −0.1 sur la note finale de projet par rendu insuffisant
  - 1: À améliorer
  - 2: Suffisant

Example

Hes·so VALAIS WALLIS
Haute Ecole d'Ingénierie
Hochschule für Ingenieurwissenschaften

- Simulation de source de courant
- Charge batterie (condo)
- Courant correct et stable (30 mA)
- Z:\I4.1-OT1\I4.12-ElnC\Project2026\Ltspice\CurrSource.asc

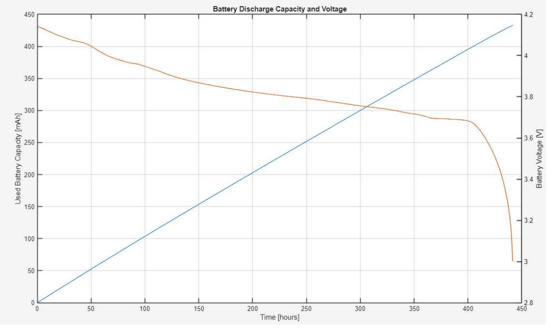




Example

Hes·so VALAIS WALLIS
Haute Ecole d'Ingénierie
Hochschule für Ingenieurwissenschaften

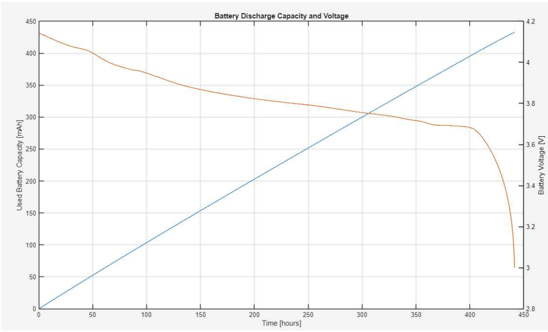
- Mesure de la capacité de la batterie et sa tension
- Importation et représentation graphique des données :
  - Z:\I4.1-OT1\I4.12-ElnC\Project2026\MATLAB\PlotBatteryCapacity.m
- Données dans le fichier :
  - Z:\I4.1-OT1\I4.12-ElnC\Project2026\data\BatteryCapacityMesurements.csv



Bad Plot! Bad graphics!

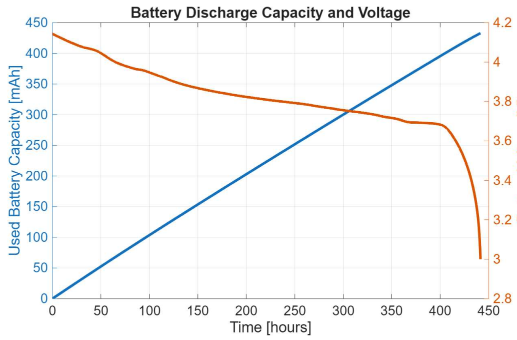
- For your final rapport you must produce high quality graphics
- No screenshots form the oscilloscope!
- Use provided MATLAB files to download data and plot it
- Graphics must be relevant and informative; they are not decoration!

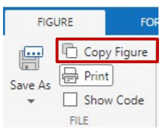
What is wrong?

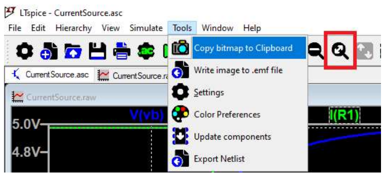


Good Plot! Good Graphics!

- Text is sufficiently large to read
- Lines are thick enough
- Colours of axes and plot lines go together making it clear which line goes with which axis.
- **Image exporter** gives a clean result! Don't use screenshots or snipping tool when there is an alternative!
- DOES IT LOOK GOOD?
- DOES IT HAVE INFORMATION?







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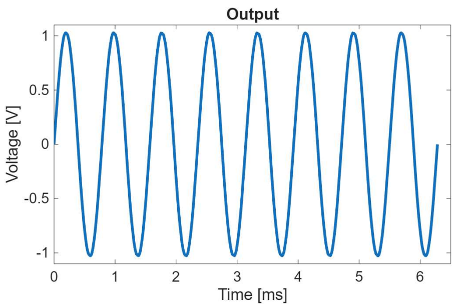

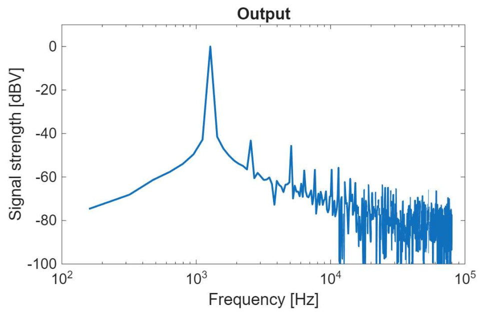

Why are you plotting a sinus???
Just say that you measured a sinus.
The picture adds no information!
Graphics should not be PROOF OF WORK!

Spectrum of sinus gives information!
How good a sinus is it?
We see harmonics and noise at higher frequencies
I should probably quantify this also

Writing project specifications (cahier de charge)

- 15 minutes to read about batteries (Lithium Ion 18650)
- 30 minutes collective brainstorming
- 30 minutes to read more and make a quick draft specification
- 30 minutes collective outlining of required features
- 1 hour to finish your draft specification document
  - Block diagram (hardware and software)
  - State diagram (hardware and software)
  - Sequence diagram
  - No circuits yet – functional blocks!
  - Supporting documents